Decision Trees, Ensembles and Validation

LumoraAI Presentation Demo • 04 Machine Learning

Decision trees

• A decision tree recursively partitions data using feature-based conditions. The split criterion can be based on measures such as Gini impurity or entropy for classification.

Ensembles

• Random forests average or vote across many decision trees trained with randomized sampling and feature selection. Gradient boosting builds learners sequentially so that later learners focus on correcting earlier errors.

Validation

• Cross-validation repeatedly trains and evaluates a model on different train-validation splits. It provides a more reliable estimate of generalization than evaluating on a single training split.

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